

Geometry
Unit 7
Lesson 5 Practice

Name _____
Date _____
Period _____

$\triangle PGL$ is a right triangle where $m\angle LPG = 90^\circ$, $LP = 8$, and $LG = 17$.

1. Determine $m\angle LGP$ using a sine ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$\triangle BLC$ is a right triangle where $m\angle BCL = 90^\circ$, $BC = 33$, and $LC = 56$.

2. Determine $m\angle CBL$ using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$\triangle MDZ$ is a right triangle where $m\angle DZM = 90^\circ$, $MD = 41$, and $DZ = 9$.

3. Determine the length of $\overline{ZM}$. If necessary, round your answer to the nearest thousandth.
4. Determine $m\angle MDZ$ using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$\triangle TKB$ is a right triangle where $m\angle KTB = 90^\circ$, $KT = 92$, and $KB = 115$.

5. Determine the length of $\overline{TB}$. If necessary, round your answer to the nearest thousandth.

6. Determine $m\angle TKB$ using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$\triangle SPC$ is a right triangle where $m\angle CSP = 90^\circ$, $m\angle CPS = 27^\circ$, and $CS = 14$.

7. Determine the length of $\overline{CP}$ using a sine ratio. Show your work. If necessary, round your answer to the nearest thousandth.

8. Explain why the tangent ratio cannot be used to determine the length of $\overline{CP}$.

$\triangle NLT$ is a right triangle where $m\angle NTL = 90^\circ$, $m\angle TNL = 33^\circ$, and $NL = 157$.

9. Determine the length of $\overline{TL}$. Show your work. If necessary, round your answer to the nearest thousandth.

10. Determine the length of $\overline{NT}$. Show your work. If necessary, round your answer to the nearest thousandth.